Math 110 — Week 6

Invariant subspaces, eigenvalues and eigenvectors, PageRank (Notes Ch. 11, LADR 5A, HW6)

§1 Facts used from Weeks 1–5

(W1) In a space of dimension n, any $n+1$ vectors are dependent, and an independent list has at most n vectors (Week 2).

(W2) Dependence lemma: in a dependent list $v_1, \dots, v_m$ some v_k lies in $\operatorname{span}(v_1, \dots, v_{k-1})$ (Week 2).

(W3) For an operator T on a finite-dimensional space: injective $\iff$ surjective $\iff$ invertible. T is injective $\iff \operatorname{null} T = \{0\}$ (Weeks 3, 4).

(W4) $A = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$ is invertible $\iff \det A = ad - bc \neq 0$ (Week 4).

(W5) $\mathcal{M}(T)\mathcal{M}(v) = \mathcal{M}(Tv)$: applying T multiplies the coordinate column by the matrix (Week 3).

(W6) A stochastic matrix P has entries ≥ 0 and each column sums to 1. If p is a probability vector, so is Pp (Week 3).

(W7) If $T(U_1) \subseteq U_2$, then $v + U_1 \mapsto Tv + U_2$ is a well-defined linear map $V/U_1 \to W/U_2$ (Week 5).

§2 This week

§2.1 Invariant subspaces, eigenvectors, eigenspaces (Ch. 11.1)

The question. An operator $T\colon V \to V$ maps a space to itself, so it can be applied repeatedly. To understand T we try to cut V into smaller pieces that T maps into themselves, and study T on each piece. What are the smallest such pieces?

Definition (Def. 30 / LADR 5.2)**.** Let $T \in \mathcal{L}(V)$. A subspace $U \subseteq V$ is **invariant under** T if $Tu \in U$ for every $u \in U$.

- To check invariance it is enough to check $Tu \in U$ for u in a spanning list of U.
- If U is invariant, T gives two smaller operators: the **restriction** $T|_U\colon U \to U$, and the **quotient operator** $T/U\colon V/U \to V/U$, $(T/U)(v+U) = Tv + U$ (well-defined by (W7) with $U_1 = U_2 = U$).
- Ex. 48: $\{0\}$, V, $\operatorname{null} T$ and $\operatorname{range} T$ are always invariant. ($u \in \operatorname{null} T \Rightarrow Tu = 0 \in \operatorname{null} T$. $u \in \operatorname{range} T \Rightarrow Tu \in \operatorname{range} T$ because Tu is an output of T.)

Definition (Def. 31, 32 / LADR 5.5, 5.8, 5.52)**.** A *nonzero* vector v is an **eigenvector** of T if $Tv = \lambda v$ for some $\lambda \in \mathbb{F}$. The scalar λ is the **eigenvalue** of v. The **eigenspace** of λ is

$$E(\lambda, T) = \operatorname{null}(T - \lambda I) = \{v : Tv = \lambda v\}.$$

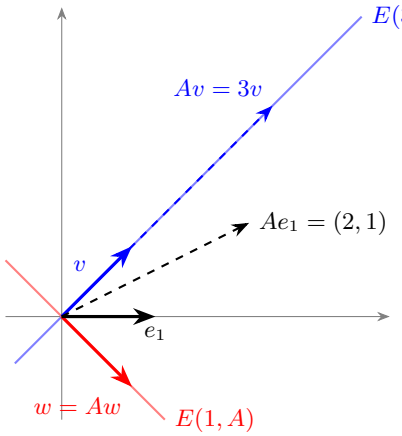
- **An eigenvector is the same as a one-dimensional invariant subspace.** If $\operatorname{span}(v)$ is invariant, then $Tv \in \operatorname{span}(v)$, so $Tv = \lambda v$. Conversely if $Tv = \lambda v$ then $T(cv) = c\lambda v \in \operatorname{span}(v)$.
- λ is determined by v: if $\lambda v = \mu v$ with $v \neq 0$ then $\lambda = \mu$.
- $E(\lambda, T)$ consists of 0 and all eigenvectors with eigenvalue λ. It is a subspace (a null space), and it is invariant: if $Tv = \lambda v$ then $T(Tv) = \lambda(Tv)$.
- λ is an eigenvalue of T exactly when $E(\lambda, T) \neq \{0\}$.

Proposition (LADR 5.7: four ways to say "λ is an eigenvalue")

Let V be finite-dimensional. The following are equivalent: (a) λ is an eigenvalue of T; (b) $T - \lambda I$ is not injective; (c) $T - \lambda I$ is not surjective; (d) $T - \lambda I$ is not invertible.

Proof. $Tv = \lambda v$ is the same equation as $(T - \lambda I)v = 0$. So (a) says $\operatorname{null}(T - \lambda I) \neq \{0\}$, which is (b). For an operator on a finite-dimensional space, (b), (c), (d) are equivalent (W3). □

In particular: 0 **is an eigenvalue of** $T \iff T$ **is not invertible**, and $E(0, T) = \operatorname{null} T$.



$A = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right)$.
$v = (1, 1)$: $Av = (3, 3) = 3v$. The line through v is mapped into itself.
$w = (1, -1)$: $Aw = (1, -1) = w$.
$e_1 = (1, 0)$: $Ae_1 = (2, 1)$ is not a multiple of e_1. The line through e_1 is not invariant.

Eigenvectors are the vectors whose direction is kept. Solid arrows are vectors, dashed arrows their images.

Example (Ex. 49: all invariant subspaces of three operators on $\mathbb{R}^2$)

In $\mathbb{R}^2$ the subspaces are $\{0\}$, lines, and $\mathbb{R}^2$. An invariant line is $\operatorname{span}(v)$ with v an eigenvector. So finding invariant subspaces means finding eigenvectors.

1. $\left(\begin{smallmatrix} 1 & 0 \\ 0 & 2 \end{smallmatrix}\right)$: $(x, y) \mapsto (x, 2y)$ is a multiple of (x, y) only if $x = 0$ or $y = 0$. Invariant: $\{0\}$, $\operatorname{span}(e_1)$, $\operatorname{span}(e_2)$, $\mathbb{R}^2$.
2. $\left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$ (rotation by 90°): no nonzero vector is mapped to a multiple of itself. Invariant: only $\{0\}$ and $\mathbb{R}^2$.
3. $N = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$: $N(x, y) = (y, 0) = \lambda(x, y)$ forces $\lambda y = 0$ and $y = \lambda x$, so $y = 0$ (and $\lambda = 0$). Invariant: $\{0\}$, $\operatorname{span}(e_1)$, $\mathbb{R}^2$. The line $\operatorname{span}(e_2)$ is not invariant because $Ne_2 = e_1$.

For N the chain $\{0\} \subset \operatorname{span}(e_1) \subset \mathbb{R}^2$ consists of invariant subspaces. Chains of invariant subspaces $\operatorname{span}(v_1) \subset \operatorname{span}(v_1, v_2) \subset \dots$ are what Week 7 uses to bring a matrix to triangular form.

Stationary distributions are eigenvectors. For the stochastic matrix of Week 3,

$$P = \begin{pmatrix} 1/2 & 1/4 & 0 \\ 1/2 & 1/2 & 1/2 \\ 0 & 1/4 & 1/2 \end{pmatrix},$$

the probability vector $\pi = (\frac{1}{4}, \frac{1}{2}, \frac{1}{4})$ satisfies $P\pi = \pi$: it is an eigenvector with eigenvalue 1. A probability vector with $P\pi = \pi$ is called a **stationary distribution**; then $P^k\pi = \pi$ for all k. Four separate questions: does one exist, is it unique, does $P^k p_0$ converge to it, and how fast.

§2.2 Complex numbers and the Fundamental Theorem of Algebra (Ch. 11.2)

The question. Does every operator have an eigenvector? Rotation of $\mathbb{R}^2$ by 90° has none. What changes over $\mathbb{C}$?

- For $z = a + bi$: $|z| = \sqrt{a^2 + b^2} = \sqrt{z\bar{z}}$, the distance from z to 0. $|zw| = |z||w|$ and $|z + w| \le |z| + |w|$.
- Polar form: $z = r(\cos\theta + i\sin\theta)$ with $r = |z|$. De Moivre: $z^k = r^k(\cos k\theta + i\sin k\theta)$. So every nonzero z has a k-th root:

$$r^{1/k}\big(\cos\frac{\theta}{k} + i\sin\frac{\theta}{k}\big).$$

Theorem (Prop. 52: Fundamental Theorem of Algebra)
Every nonconstant polynomial with complex coefficients has a zero in $\mathbb{C}$.

The idea

Look at the size $|p(z)|$. Far from 0 it is large, so it has a smallest value somewhere, at a point ζ. We want to show that smallest value is 0. Suppose it is not. Then we will find a nearby point where $|p|$ is even smaller, a contradiction.

Why can we always go lower from a nonzero value? Try $q(z) = 1 + z^2$ at $z = 0$, where $|q| = 1$. Moving along the real axis, $q(t) = 1 + t^2$ gets bigger. Moving along the imaginary axis, $q(it) = 1 - t^2$ gets smaller. The term z^2 can be made to point in any direction by choosing the direction of z; choose it so that z^2 points opposite to the constant 1.

In general, after moving ζ to the origin and dividing by $p(\zeta)$, the polynomial is $q(z) = 1 + a_k z^k +$ (higher powers) with $a_k \neq 0$. Choose the direction β of z so that $a_k\beta^k = -1$ (possible because k-th roots exist in $\mathbb{C}$). Along $z = t\beta$ the polynomial is $1 - t^k +$ (terms of size about t^{k+1}), and for small t the term $-t^k$ wins.

Proof. Let $p(z) = c_0 + c_1 z + \cdots + c_m z^m$ with $m \ge 1$, $c_m \neq 0$.

1. $|p(z)|$ *is large far away.* By the triangle inequality $|p(z)| \ge |c_m||z|^m - \sum_{j<m} |c_j||z|^j$, and the right side tends to ∞ as $|z| \to \infty$. So there is a radius R with $|p(z)| > |p(0)|$ whenever $|z| > R$.
2. *A global minimum exists.* On the closed disk $|z| \le R$ the continuous real function $|p|$ attains a minimum, at some ζ. That minimum is $\le |p(0)|$, and outside the disk $|p|$ is larger than $|p(0)|$. So $|p(\zeta)| \le |p(z)|$ for all $z \in \mathbb{C}$.
3. *Suppose* $p(\zeta) \neq 0$. Let $q(z) = p(\zeta + z)/p(\zeta)$. Then $q(0) = 1$ and $|q(z)| \ge 1$ for all z.
4. q is a nonconstant polynomial, so $q(z) = 1 + a_k z^k + a_{k+1} z^{k+1} + \cdots + a_m z^m$ where $a_k z^k$ is the first nonzero term after the constant.
5. Choose $\beta \in \mathbb{C}$ with $\beta^k = -1/a_k$. For real $t > 0$: $q(t\beta) = 1 - t^k + a_{k+1}\beta^{k+1}t^{k+1} + \cdots + a_m\beta^m t^m$.
6. For $0 < t < 1$ the tail has absolute value at most Ct^{k+1}, where $C = \sum_{j>k} |a_j\beta^j|$. So

$$|q(t\beta)| \le (1 - t^k) + Ct^{k+1} = 1 - t^k(1 - Ct).$$

7. For t small enough that $Ct < 1$, this is < 1. That contradicts $|q| \ge 1$. Hence $p(\zeta) = 0$.

□

- **Factorisation.** If $p(\lambda) = 0$ then $p(z) = (z - \lambda)q(z)$ with $\deg q = \deg p - 1$. Applying the theorem repeatedly: $p(z) = c(z - \lambda_1)\cdots(z - \lambda_m)$, roots listed with multiplicity. $\mathbb{C}$ is called **algebraically closed**.
- $\mathbb{R}$ is not: $x^2 + 1$ has no real zero.
- The only analytic input is that a continuous function on a closed bounded set attains its minimum.

§2.3 Existence and independence of eigenvectors (Ch. 11.2)

Theorem (Prop. 53 / LADR 5.19: operators on complex spaces have eigenvectors)
Every operator on a nonzero finite-dimensional complex vector space has an eigenvector.

The idea

We need a polynomial so that the Fundamental Theorem can be used, but so far T has no polynomial attached to it. Finite dimension produces one. Take any $v \neq 0$ and keep applying T: the vectors $v, Tv, T^2v, \dots, T^nv$ are $n+1$ vectors in a space of dimension n, so they satisfy a relation $a_0v + a_1Tv + \cdots + a_mT^mv = 0$. Read that as "$p(T)v = 0$" for the polynomial $p(z) = a_0 + a_1z + \cdots + a_mz^m$.

Now factor: $p(z) = (z-\lambda)q(z)$. Then $(T-\lambda I)\big(q(T)v\big) = p(T)v = 0$. So $w = q(T)v$ satisfies $Tw = \lambda w$. It is an eigenvector provided $w \neq 0$. To guarantee that, take the relation of *smallest* degree at the start: q has smaller degree than p, so $q(T)v = 0$ would be a shorter relation.

Example: T = rotation by 90° on $\mathbb{C}^2$, $v = e_1$. $Te_1 = e_2$, $T^2e_1 = -e_1$, so $T^2v + v = 0$: $p(z) = z^2+1 = (z-i)(z+i)$. Take $\lambda = i$, $q(z) = z+i$: $w = Te_1 + ie_1 = (i, 1)$, and $Tw = (-1, i) = i \cdot (i, 1)$.

Proof. Let $n = \dim V$ and choose $v \neq 0$.

1. $v, Tv, \dots, T^nv$ are $n+1$ vectors, so they are dependent (W1). Among all nontrivial relations $a_0v + a_1Tv + \cdots + a_mT^mv = 0$ ($a_m \neq 0$) choose one with m as small as possible. Then $m \geq 1$, because $a_0v = 0$ with $a_0 \neq 0$ is impossible.
2. Let $p(z) = a_0 + a_1z + \cdots + a_mz^m$. It has a zero $\lambda \in \mathbb{C}$, so $p(z) = (z-\lambda)q(z)$ with $q(z) = b_0 + \cdots + b_{m-1}z^{m-1}$, $b_{m-1} = a_m \neq 0$.
3. Let $w = b_0v + b_1Tv + \cdots + b_{m-1}T^{m-1}v$. Then $w \neq 0$: otherwise this would be a nontrivial relation of degree $m-1 < m$.
4. Multiplying out $(z-\lambda)q(z) = p(z)$ with T in place of z:
$$(T-\lambda I)w = a_0v + a_1Tv + \cdots + a_mT^mv = 0.$$
5. So $Tw = \lambda w$ with $w \neq 0$.

□

- Complex is needed: rotation by 90° on $\mathbb{R}^2$. Finite-dimensional is needed: $T(p) = xp$ on $\mathcal{P}(\mathbb{C})$ has no eigenvector, since $xp = \lambda p$ fails by comparing degrees.

Theorem (Prop. 54 / LADR 5.11: eigenvectors for distinct eigenvalues are independent)
If $v_1, \dots, v_m$ are eigenvectors of T with distinct eigenvalues $\lambda_1, \dots, \lambda_m$, then $v_1, \dots, v_m$ is linearly independent.

The idea

Start with two. Could $v_2 = av_1$? Apply T to both sides: $\lambda_2v_2 = a\lambda_1v_1$. Multiply both sides by λ_2 instead: $\lambda_2v_2 = a\lambda_2v_1$. The left sides are equal, so $a(\lambda_1 - \lambda_2)v_1 = 0$, so $a = 0$, so $v_2 = 0$, which an eigenvector is not.

The trick: *applying T and multiplying by λ_k do the same thing to v_k but different things to the other eigenvectors.* Subtracting removes v_k and leaves a relation among fewer vectors, with each coefficient multiplied by a nonzero number $\lambda_i - \lambda_k$. For the general case we want the shorter relation to be among independent vectors; so pick the *first* v_k that depends on the earlier ones.

Proof. 1. Suppose the list is dependent. By (W2) there is a smallest k with $v_k \in \operatorname{span}(v_1, \dots, v_{k-1})$. Because k is smallest, $v_1, \dots, v_{k-1}$ is independent.

2. Write $v_k = a_1v_1 + \cdots + a_{k-1}v_{k-1}$.
3. Apply T: $\lambda_kv_k = a_1\lambda_1v_1 + \cdots + a_{k-1}\lambda_{k-1}v_{k-1}$.
4. Multiply step 2 by λ_k: $\lambda_kv_k = a_1\lambda_kv_1 + \cdots + a_{k-1}\lambda_kv_{k-1}$.
5. Subtract: $0 = a_1(\lambda_1 - \lambda_k)v_1 + \cdots + a_{k-1}(\lambda_{k-1} - \lambda_k)v_{k-1}$.

6. $v_1, \dots, v_{k-1}$ is independent, so each $a_i(\lambda_i - \lambda_k) = 0$. The eigenvalues are distinct, so each $a_i = 0$.
7. Then $v_k = 0$ by step 2, contradicting that eigenvectors are nonzero.

□

- **Prop. 55 / LADR 5.12.** An operator on a finite-dimensional V has at most $\dim V$ distinct eigenvalues: eigenvectors for distinct eigenvalues are independent, and an independent list has at most $\dim V$ vectors (W1).
- Ex. 50: $\sin x, \sin 2x, \dots, \sin Nx$ are independent, because they are eigenvectors of $f \mapsto f''$ with the distinct eigenvalues $-1, -4, \dots, -N^2$.
- The shift $(a_1, a_2, \dots) \mapsto (a_2, a_3, \dots)$ on $\mathbb{R}^\infty$ has every real λ as an eigenvalue, with eigenvector $(1, \lambda, \lambda^2, \dots)$.

§2.4 Computing eigenvalues (Ch. 11.2)

Proposition (Prop. 56: eigenvalues of an operator are eigenvalues of its matrix)
Fix a basis of V. Then

$$Tv = \lambda v \iff \mathcal{M}(T)\mathcal{M}(v) = \lambda \mathcal{M}(v).$$

Proof. $\mathcal{M}(T)\mathcal{M}(v) = \mathcal{M}(Tv)$ (W5) and $\mathcal{M}(\lambda v) = \lambda \mathcal{M}(v)$; two vectors are equal exactly when their coordinate columns are equal. □

The 2×2 characteristic polynomial. For $A = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$:

1. λ is an eigenvalue $\iff$ $\lambda I - A$ is not invertible $\iff$ $\det(\lambda I - A) = 0$ (W4).
2. Apply the 2×2 formula to $tI - A$:

   $$\det(tI - A) = (t-a)(t-d) - bc = t^2 - (a+d)t + (ad - bc).$$

   Call it $\chi_A(t)$.
3. So the eigenvalues of A are the zeros of χ_A in $\mathbb{F}$. Their sum is $a + d$ and their product is $ad - bc$.
4. Eigenvectors for λ: solve $(A - \lambda I)x = 0$. The two rows are proportional, so use one: $(a - \lambda)x + by = 0$ gives $(b,\ \lambda - a)$; or $cx + (d - \lambda)y = 0$ gives $(\lambda - d,\ c)$. Use whichever is nonzero.

(Only the 2×2 formula is available for now; the characteristic polynomial in general comes later in the course.)

- Ex. 51: the rotation R and the Fibonacci matrix F have characteristic polynomials

  $$R = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad \chi_R = t^2 + 1; \qquad F = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}, \quad \chi_F = t^2 - t - 1.$$

  So R has no real eigenvalues, and over $\mathbb{C}$ its eigenvalues are $\pm i$. The eigenvalues of F are $(1 \pm \sqrt{5})/2$.
- **Ex. 52: use the structure, not a determinant.** $T(A) = A^\mathsf{T}$ on $\mathbb{C}^{2,2}$ satisfies $T^2 = I$. If $TA = \lambda A$ with $A \neq 0$, then $A = T^2 A = \lambda^2 A$, so $\lambda^2 = 1$. $E(1, T)$ is the symmetric matrices, spanned by $\left(\begin{smallmatrix} 1 & 0 \\ 0 & 0 \end{smallmatrix}\right), \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right), \left(\begin{smallmatrix} 0 & 0 \\ 0 & 1 \end{smallmatrix}\right)$; $E(-1, T)$ is spanned by $\left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$.

§2.5 PageRank (Ch. 11.3)

The question. Rank n web pages by importance using only the links, where a page is important if important pages link to it.

Setting up.

- Adjacency matrix: $C_{ij} = 1$ if page i links to page j, else 0. $(C^k)_{ij}$ counts directed walks of length k from i to j.
- Let d_j be the number of links out of page j (assume $d_j > 0$). Define $P_{ij} = 1/d_j$ if page j links to page i, else 0. Column j sums to 1, so P is stochastic. In matrix form $P = C^\mathsf{T} D^{-1}$ with $D = \operatorname{diag}(d_1, \dots, d_n)$.
- A "random surfer" on page j follows one of its links uniformly at random; if p is the distribution of the surfer's position, Pp is the distribution one step later.

- First attempt: rank by a stationary distribution $\pi = P\pi$. Coordinate i says $\pi_i = \sum_j P_{ij}\pi_j$: the importance of page i is collected from the pages linking to it.
- Problems: a page with no outgoing links (a **dangling page**) gives a zero column; the stationary distribution may not be unique; $P^k p_0$ may not converge. Dangling columns are replaced by the vector v below.

Small example: pages $1, 2, 3$ with links $1 \to 2$, $1 \to 3$, $2 \to 3$, $3 \to 1$.

$$P = \begin{pmatrix} 0 & 0 & 1 \\ 1/2 & 0 & 0 \\ 1/2 & 1 & 0 \end{pmatrix}.$$

Solving $P\pi = \pi$: $\pi_1 = \pi_3$, $\pi_2 = \frac{1}{2}\pi_1$, so $\pi = (\frac{2}{5}, \frac{1}{5}, \frac{2}{5})$. **The PageRank equation.** Fix a **damping parameter** $0 < \alpha < 1$ and a probability vector v. At each step the surfer follows a link with probability α, and with probability $1 - \alpha$ jumps to a page chosen according to v. The PageRank vector r is the solution of

$$r = \alpha P r + (1 - \alpha)v, \qquad \text{equivalently} \qquad (I - \alpha P)r = (1 - \alpha)v.$$

Proposition (Prop. 57: eigenvalues of a stochastic matrix have absolute value at most 1)

The idea

A stochastic matrix redistributes mass without creating any. Measure the total mass of a (possibly complex) vector by $\|x\|_1 = |x_1| + \cdots + |x_n|$. Then $\|Px\|_1 \le \|x\|_1$. An eigenvector is scaled by $|\lambda|$, so $|\lambda| \le 1$.

Proof. 1. By the triangle inequality and $P_{ij} \ge 0$,

$$\|Px\|_1 = \sum_i \Big|\sum_j P_{ij}x_j\Big| \le \sum_i \sum_j P_{ij}|x_j|.$$

2. Exchange the sums:

$$\sum_j \Big(\sum_i P_{ij}\Big)|x_j| = \sum_j |x_j| = \|x\|_1,$$

because each column sums to 1.

3. If $Px = \lambda x$ with $x \neq 0$: $|\lambda|\,\|x\|_1 = \|Px\|_1 \le \|x\|_1$, and $\|x\|_1 > 0$, so $|\lambda| \le 1$.

□

Theorem (Prop. 58: existence and uniqueness of PageRank)

For a stochastic P, a probability vector v and $0 < \alpha < 1$, there is exactly one $r \in \mathbb{R}^n$ with $r = \alpha P r + (1 - \alpha)v$, and r is a probability vector.

The idea

The equation is the linear system $(I - \alpha P)r = (1 - \alpha)v$, so everything follows if $I - \alpha P$ is invertible. For a square matrix it is enough to show injectivity (W3). And $(I - \alpha P)x = 0$ says $Px = \frac{1}{\alpha}x$: that would make $\frac{1}{\alpha} > 1$ an eigenvalue of P, which Prop. 57 forbids. Uniqueness gives existence.

To see that r is a probability vector, write the solution explicitly. Substituting the equation into itself:

$$r = (1 - \alpha)v + \alpha P r = (1 - \alpha)v + \alpha(1 - \alpha)Pv + \alpha^2 P^2 r = \ldots,$$

which suggests $r = (1 - \alpha)\sum_{k \ge 0} \alpha^k P^k v$. This is a weighted average of the probability vectors $P^k v$ with weights $(1 - \alpha)\alpha^k$ that sum to 1.

Proof. 1. *$I - \alpha P$ is injective.* If $(I - \alpha P)x = 0$ with $x \neq 0$, then $Px = \frac{1}{\alpha}x$, so $\frac{1}{\alpha} > 1$ is an eigenvalue of P, contradicting Prop. 57.

2. *Unique solution.* An injective operator on $\mathbb{R}^n$ is invertible (W3). So $(I - \alpha P)r = (1 - \alpha)v$ has exactly one solution.

3. *The series.* Let
$$s = (1-\alpha)\sum_{k=0}^{\infty} \alpha^k P^k v.$$
Each $P^k v$ is a probability vector (W6), so each coordinate of each term lies in $[0, (1-\alpha)\alpha^k]$ and the series converges by comparison with a geometric series. The coordinates of s are ≥ 0.

4. *Sum of coordinates.*
$$\mathbf{1}^\mathsf{T} s = (1-\alpha)\sum_k \alpha^k \mathbf{1}^\mathsf{T} P^k v = (1-\alpha)\sum_k \alpha^k = 1.$$
So s is a probability vector.

5. *s solves the equation.*
$$\alpha P s + (1-\alpha)v = (1-\alpha)\sum_{k\geq 1} \alpha^k P^k v + (1-\alpha)v = s.$$
By uniqueness $s = r$.

□

Computing r by iteration. Start from any probability vector r_0 and set $r_{k+1} = \alpha P r_k + (1-\alpha)v$.

1. Subtract $r = \alpha P r + (1-\alpha)v$: $r_{k+1} - r = \alpha P(r_k - r)$.
2. By step 1–2 of Prop. 57, $\|r_{k+1} - r\|_1 \leq \alpha \|r_k - r\|_1$. So $\|r_k - r\|_1 \leq \alpha^k \|r_0 - r\|_1 \to 0$: the error shrinks by a factor α each step.
3. The map $F(x) = \alpha P x + (1-\alpha)v$ satisfies $\|F(x) - F(y)\|_1 \leq \alpha \|x - y\|_1$; such a map is a **contraction** (Banach fixed point theorem in analysis).

The Google matrix. Let $\mathbf{1} = (1, \ldots, 1)^\mathsf{T}$ and $G = \alpha P + (1-\alpha)v\mathbf{1}^\mathsf{T}$.

- For a probability vector x, $\mathbf{1}^\mathsf{T} x = 1$, so $Gx = \alpha P x + (1-\alpha)v$. The PageRank equation is $Gr = r$: r is an eigenvector of G with eigenvalue 1.
- Let $H = \{x \in \mathbb{R}^n : x_1 + \cdots + x_n = 0\}$. P keeps the sum of coordinates, so H is invariant under P. For $x \in H$, $\mathbf{1}^\mathsf{T} x = 0$ and $Gx = \alpha P x \in H$. So H is invariant under G, and on H the matrix G acts as αP.
- $\mathbb{R}^n = \operatorname{span}(r) \oplus H$: $r \notin H$ since its coordinates sum to 1, and $\dim H = n - 1$. So G fixes the direction r and shrinks the invariant complement H by at least the factor α.
- History: Brin and Page used $\alpha = 0.85$ and uniform v; a non-uniform v gives personalised PageRank.

Shortcuts

- 2×2: $\chi_A(t) = t^2 - (a+d)t + (ad - bc)$. Eigenvector for λ: $(b, \lambda - a)$ or $(\lambda - d, c)$.
- **Polynomial equation for T gives the candidates**: if $Tv = \lambda v$ then $T^k v = \lambda^k v$, so $p(T)v = p(\lambda)v$. If $p(T) = 0$, every eigenvalue is a root of p. ($P^2 = P$: $\lambda \in \{0, 1\}$. $T^2 = I$: $\lambda = \pm 1$. $N^2 = 0$: $\lambda = 0$.)
- **n distinct eigenvalues in dimension n**: the eigenvectors form a basis (Prop. 54 and counting). No further check needed.
- **0 is an eigenvalue $\iff$ not invertible.**
- **Real matrix, complex eigenvalue**: if $Av = \lambda v$ then $A\bar{v} = \bar{\lambda}\bar{v}$ (take complex conjugates). Find one eigenvector and conjugate it.
- **Stochastic matrix**: 1 is always an eigenvalue. Reason: $P - I$ has columns summing to 0, so its range lies in H, so it is not surjective, so it is not injective (W3). If P is also symmetric (rows sum to 1 too), the uniform vector is stationary.
- **Conjugate operators**: $S^{-1}TS$ has the same eigenvalues as T, with eigenspaces $S^{-1}E(\lambda, T)$ (HW6 P3).
- **Degree arguments on polynomial spaces**: differentiation lowers degree, so $p' = \lambda p$ forces $\lambda = 0$ (HW6 P2).
- **Invariant subspace check**: apply T to a spanning list.

Traps

- An eigenvector is nonzero by definition; an eigenvalue may be 0.
- The sum of two eigenvectors with different eigenvalues is not an eigenvector. (Within one eigenspace, sums are fine.)
- Over $\mathbb{R}$ an operator may have no eigenvalue. In infinite dimensions an operator may have none even over $\mathbb{C}$.
- Independence needs *distinct* eigenvalues. Two eigenvectors with the same eigenvalue can be dependent.
- $\operatorname{null} T$ and $\operatorname{range} T$ are invariant, but an invariant subspace need not have an invariant complement: for N, the only invariant line is $\operatorname{span}(e_1)$.
- $|\lambda| \le 1$ for stochastic matrices is about absolute values: $\lambda = -1$ or complex λ with $|\lambda| = 1$ can occur (permutation matrices).

§3 Problem set

HW6 P1. Let $P\colon V \to V$ satisfy $P^2 = P$. (a) Prove that every eigenvalue of P belongs to $\{0, 1\}$. (b) Prove that $V = \operatorname{null} P \oplus \operatorname{range} P$.

Hint: (a) apply P twice to an eigenvector. (b) $v = (v - Pv) + Pv$.

HW6 P2. Let $D\colon \mathcal{P}_n(\mathbb{F}) \to \mathcal{P}_n(\mathbb{F})$ be differentiation, $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$. (a) Determine all eigenvalues and eigenspaces of D. (b) Explain why the answer is different on the space of smooth functions on $\mathbb{R}$.

Hint: compare degrees in $p' = \lambda p$.

HW6 P3. Let $S\colon V \to V$ be invertible and $R = S^{-1}TS$. Prove that T and R have the same eigenvalues; more precisely $E(\lambda, R) = S^{-1}E(\lambda, T)$ for every λ.

Hint: rewrite $Rv = \lambda v$ as a statement about Sv.

HW6 P4. Let $A = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$. Using only the 2×2 determinant formula, prove $\det(tI - A) = t^2 - (a+d)t + (ad - bc)$. Then prove that λ is an eigenvalue of A if and only if it is a root of this polynomial.

HW6 P5. Find the eigenvalues and eigenspaces over the indicated field, and decide whether there is a basis of eigenvectors: $A_1 = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$ (over $\mathbb{R}$, then over $\mathbb{C}$), $A_2 = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$ over $\mathbb{R}$, $A_3 = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 1 \end{smallmatrix}\right)$ over $\mathbb{R}$.

HW6 P6 (optional). Let $T\colon V \to V$ and $v \neq 0$. Define $C(v, T) = \operatorname{span}(v, Tv, T^2v, \dots)$. (a) Prove that $C(v, T)$ is T-invariant. (b) If V is finite-dimensional, prove there is $m \le \dim V$ with $C(v, T) = \operatorname{span}(v, Tv, \dots, T^{m-1}v)$.

Hint: (b) take the first power $T^m v$ that depends on the earlier ones.

HW6 P7 (optional). For the stochastic matrix

$$P = \begin{pmatrix} 1/2 & 1/2 & 0 \\ 1/2 & 0 & 1/2 \\ 0 & 1/2 & 1/2 \end{pmatrix}$$

find a probability vector p with $Pp = p$ and explain its meaning for the three-state Markov chain.

R1 (Midterm 2 review, Problem 3). For a polynomial p and an operator T with $Tv = \lambda v$, show $p(T)v = p(\lambda)v$.

R2 (Midterm 2 review, Problem 7). For any operator $T\colon V \to V$ prove that $\operatorname{null} T$ and $\operatorname{range} T$ are invariant under T, and that their annihilators are invariant under the dual map T'.

R3 (Midterm 2 review, true/false). (a) If v_1, v_2 are eigenvectors of T, then so is $v_1 + v_2$. (b) An operator on a real vector space always has an eigenvector.

E1 (PageRank by hand). For the three-page web in the text (links $1 \to 2$, $1 \to 3$, $2 \to 3$, $3 \to 1$) with $\alpha = \frac{1}{2}$ and uniform v, solve $r = \alpha P r + (1-\alpha)v$.